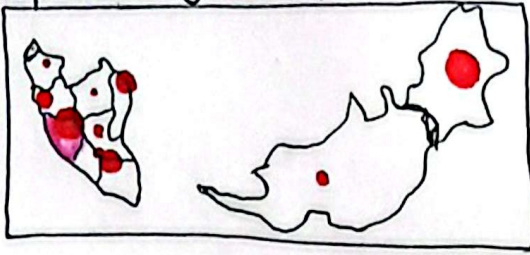
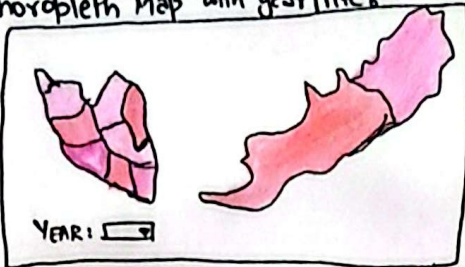


1. IDEAS

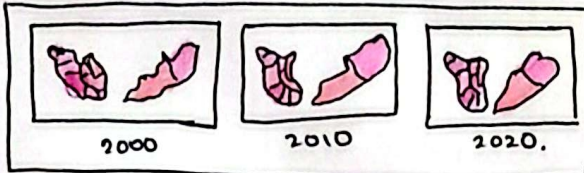
Proportional Symbol Map



Choropleth Map with year filter

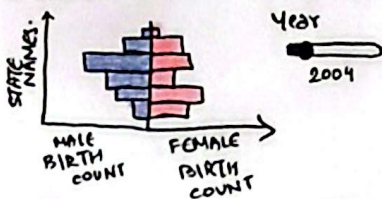


Choropleth Map only 3 yrs main



- Easy comparison

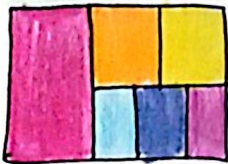
Gender & State population pyramid



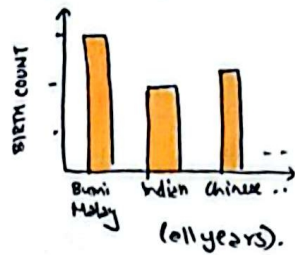
Gender Pie chart by States (Top 3)



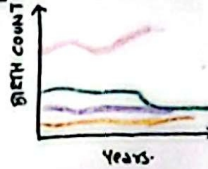
TreeMap for Ethnicity



Ethnicity Bar Chart



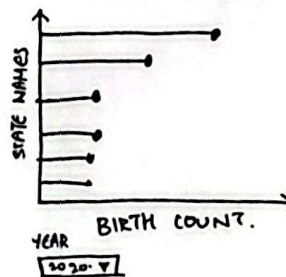
Ethnicity Line Chart



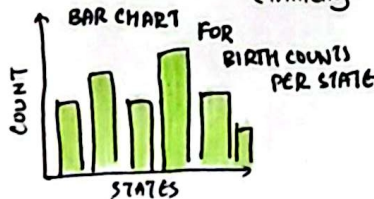
Pie Chart Ethnicity



Lollipop Chart for States (filter: Year)



Heat Map



Title: Birth Rates & Trends In Malaysia

Author: Ritika Raina

Date: 30/09/2025

Sheet: 1

Task: Brainstorming Ideas

2. FILTER

⇒ Treemap for Ethnicity can be used instead of the Ethnicity Bar Chart.

the bar chart will very plain & won't convey that much easily.

However, treemap can give us a quick glance at the ethnicities & their approx. contribution to the live births.

⇒ The bar chart & lollipop chart are almost the same for birth counts in each state. So only one should be chosen.

3. CATEGORIZE

Map view

↳ have any one map, so its good visual.

Ethnicity.

↳ have the birth count for ethnicity by years.

↳ or just like an overall view.

States

↳ how the birth counts differ in states

Gender

↳ Gender distribution across the states.

5. QUESTIONS

- Is there a flow in the visualizations?
- Are there enough visualisations?

4. COMBINE & REFINE

- States + Gender analysis
 - lollipop chart of states + population pyramid
- An overview, very basic

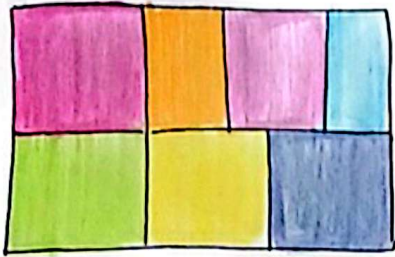
LAYOUT

image background

BIRTH RATE & TRENDS IN MY

Exploring Birth patterns, Ethnic trend & Demographic

BIRTH COUNT AND ETHNICITY

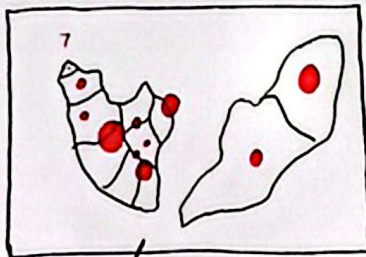


BIRTH COUNT STATE



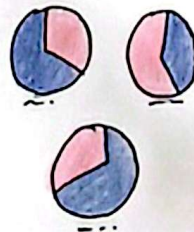
YEAR
▼

OVERVIEW OF BIRTH COUNT (2020-2023)



Proportional Symbol Map

Gender Ratio
(Top 3 states)



FOOTNOTE. (Metadata)

Title: Initial Design 1

Author: Ritika Raina

Date: 30/09/2025

Sheet: 2

Description of Task:

Making a potential dashboard design (using html).

OPERATIONS

- A year filter for the bar chart
The user selects a specific year & bar chart adjusts accordingly.
- Tooltips for maps and charts when you hover above the specific areas.
for maps: hover above the dots.

FOCUS

A general overview of the trends.

No in depth analysis.

DISCUSSION

Pros:

- good for a quick look & general information
- more visual based.

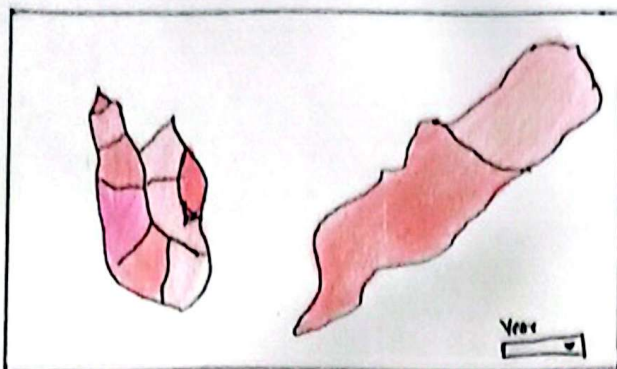
Cons:

- proportional symbol map will have the circle mark overlapping, making it messy.

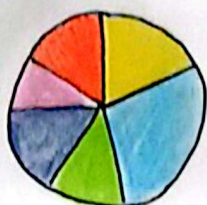
LAYOUT.

BIRTH RATE & TRENDS IN MALAYSIA

Exploring Birth Patterns, Ethnic Trend & Demographics

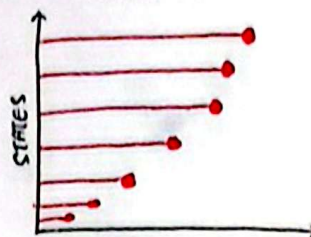


Ethnicity:



Year

BIRTH COUNT/STATE

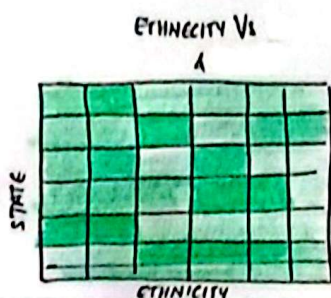


BIRTH COUNT

Year

Year

GENDER
male
female



Birth Count

FOOTNOTE (METADATA)

Title: Initial Design 2

Author: Ritika Raina

Date: 30/01/2025

Sheet 13

Description of Task:

Making a potential dashboard design.

OPERATIONS

- A year filter for each graph.
- Three of them have year sliders, so easy to see the change through the years.
- Years from 2000-2023.
- Heatmap has gender drop down selection
 - ↳ male & female options are given.
- Tooltips are present as always.

FOCUS

- A bit more detailed with year & gender option → this makes it more insightful.
- Heatmap → shows relation between the ethnicity & states giving a new perspective to the analysis. Gender wise selection can be made. Drilling down deeper.
- Choropleth map: easy to identify birth trend across the states.

DISCUSSION

Pros:

- filter → deeper insights
- slider filter
easy to see changes
- choropleth map
easier to see birth rate by state.

Cons:

- Map has 23 filter year options. makes things complicated.
- Pie chart can't show trends at once.

LAYOUT

Title: Initial Design 3

Author: Ritika Raina

Date: 30/01/2025

Sheet: 4

Description of Task:
Making a potential
dashboard layout.

OPERATIONS

- Year filters for chart (2000-2023)
- Additionally, the heatmap has gender selection drop down box.
- Tooltips are there.

DISCUSSION

Pros:

- easy comparison of birth rate using 3 separate maps.
- line chart for ethnicity makes it easier to see & follow the trend.

Cons:

Layout isn't that clear.

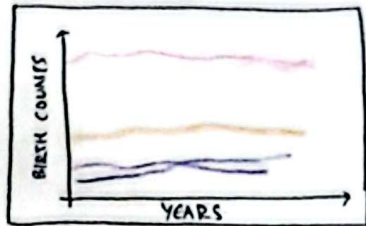
BIRTH RATE & TRENDS IN MALAYSIA

Exploring Birth patterns, ethnic trends & demographics

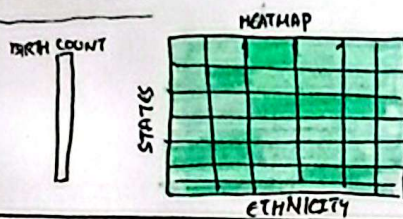
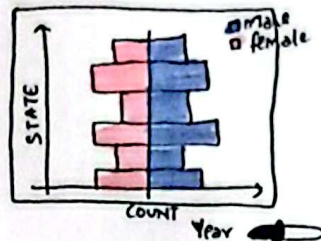
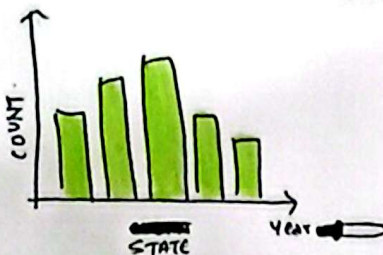
BIRTH RATE OVERVIEW



ETHNICITY BIRTH COUNTS OVER YEARS



STATES & GENDER



GENDER

FOOTNOTE (METADATA)

FOCUS

- Specific years : showing the changes by decades.
get a better view state wise & easy to compare between time periods.
- First state wise birth count, then more in detail by having a population pyramid

LAYOUT.

Light
Purple hue

Title: Final Design Sheet

Author: Ritika Raina

Date: 30/09/2025

Sheet: 5

Description of Task:
Final Implementation
of Design.

OPERATION.

- Maps giving easy comparison over decades of years. Hover above states to see more information.
- Ethnicity line chart can help for easy comparison.
- Lollipop chart overall birth count per state, population pyramid dives deeper on gender
- Heatmap gives relation between the ethnicity & states + gender views

DISCUSSION.

Next Step:

- Preparing data
- Adding lat & long values for states in the dataset.
- Make charts using VegaLite

Dependencies:

- VegaLite can make charts properly & handle filters.

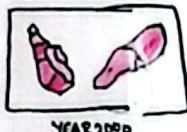
Time:

Need to work on the html file + design.

BIRTH RATE & TRENDS IN MALAYSIA

Exploring birth patterns, ethnic trends & demographic shifts

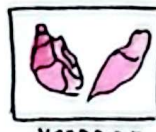
BIRTH RATE OVERVIEW



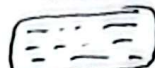
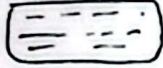
YEAR 2000



YEAR 2010

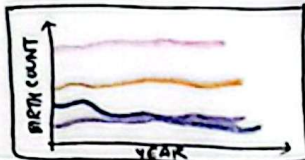


YEAR 2020



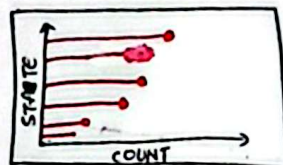
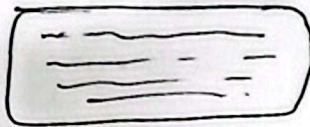
gradient colour

TREND IN BIRTH COUNT: ETHNICITY

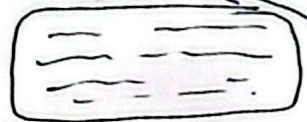
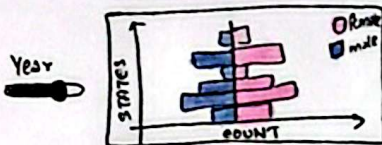


white boxes.

BIRTH TRENDS ACROSS STATES



select year



Overall BIRTH COUNTS BY GENDER & STATES



represents the birth count by ethnicity & state

FOOTNOTE (METADATA)

FOCUS

Good & detailed view of birth rate & birth counts using different attributes. Lastly combining the attributes together to help look into the relations between them.